Advantages of Using S3: -

Benefits of using S3

1. Budget-friendly

As we discussed earlier, Amazon provides services based on a pay-as-you-go model. This allows us to pay only for the amount of storage and the time that we use on S3.

Compare this with the example of a hotel room, where the cost depends on the number of rooms booked (similar to the amount of storage) and the time for which one stays (similar to the time for which one uses S3).

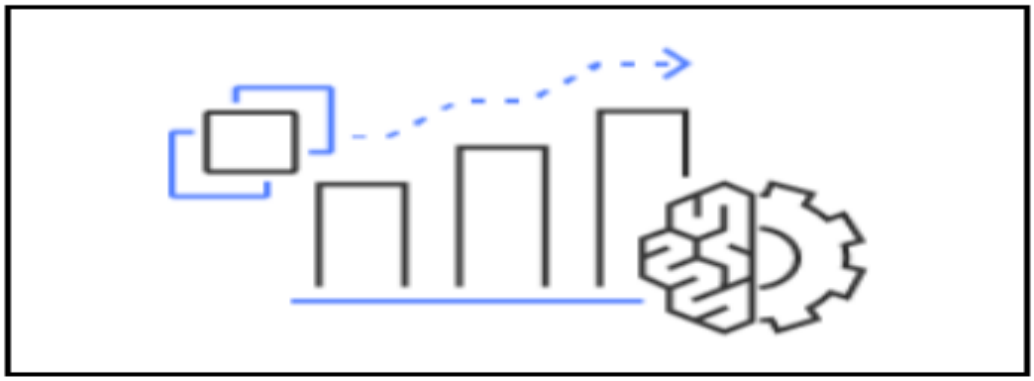
Budget-friendly AWS

2. High scalability

Scalability is the measure to increase or decrease the resource as per need.For example, you went to buy 2 chocolates from a shop, one for yourself and one for your brother. Now, when you reach home, you see that five more of your cousins have arrived. What do you do now? Obviously, sharing 2 chocolates among 7 people would not be a good idea. A better option is to buy more chocolates. Scalability in this context will mean how easily you can get more chocolates in an acceptable time.

In the age of social media, applications can get viral anytime, leading to an increase in demand for storage. But since we're using Amazon S3, we don't need to worry. S3 allows for implementation within minutes.

Moreover, it allows for **auto-scaling** as well. Be it an increase or decrease in demand, everything is perfectly managed by S3.



Auto scalable

3. Durability

Durability is the measurement of the likelihood of data loss.

All the services provided by Amazon, including Amazon S3, are highly durable.

Example

For example, you have an important document saved on your phone. You have some work and you need that document. Unluckily, you forgot your phone at your friend's house. Nothing to worry about, as you have previously saved a copy of it on your PC.

You search your computer for the file but you find out that your sibling has deleted the file! Even then, you shouldn't worry because you stored it in a cloud service as a backup. You can easily retrieve your document from there.

The S3 Standard Tier is designed for "99.999999999% durability". What does that actually mean?

It means that if we store one hundred billion objects in S3, at most we'll lose 1 object. Isn't that amazing?

4. High availability

Availability is the measure of how readily a service can be used.

For example, we want to eat a pizza from a chain of pizza restaurants. In order to proceed with the pizza order, one of the pizza chain's restaurants must be available somewhere nearby. To ensure the high availability of its products, Dominos has set up its shops in innumerable numbers across various cities, states, and countries.

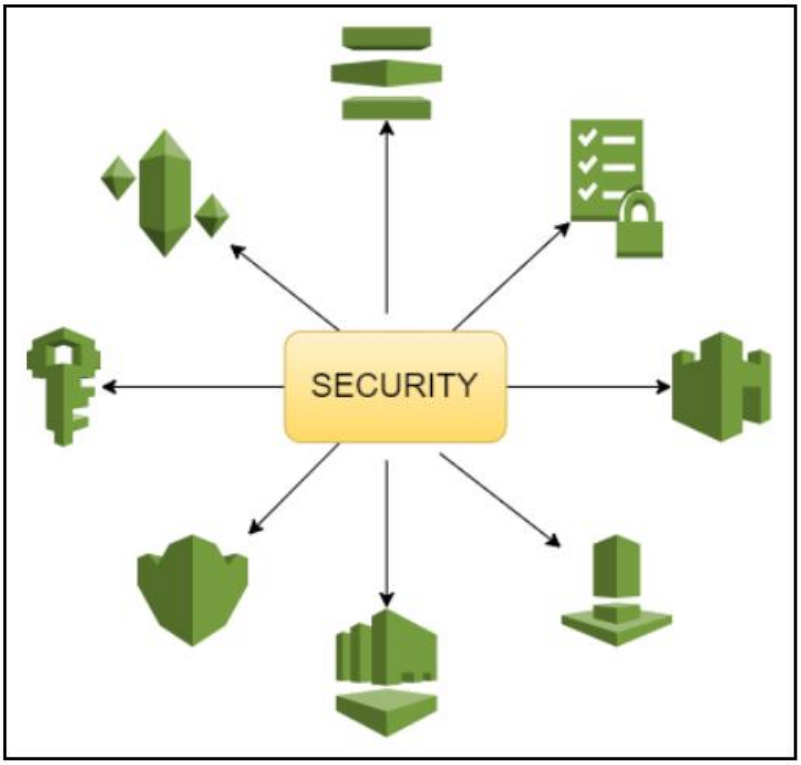
Similarly, AWS has availability zones or regions (we'll discuss these terms later in detail) across multiple countries to ensure high availability.

5. Security

When it comes to storage, the security of our data becomes the first priority. S3 relieves us by enabling automatic encryption of data as soon as the data uploading process is finished.

Various security options are available such as IAM, which allows only a certain person to access information.

In addition, we can monitor who is accessing our data, and also obtain information such as the location of access, time, and device accessing the platform.



Security

6. Easy to manage

The AWS Management Console is user-friendly. We can manage our resources in various ways.

- **GUI (Graphical User Interface):** AWS Management Console, web applications, and mobile applications, where each task can be executed with a single click or a tap.
- **CLI (Command Line Interface):** AWS provides the command line program package to execute our tasks by running the commands directly.
- **API (Application Program Interface):** AWS provides APIs which allow firms to integrate the platform with other technologies in use quickly.